

Synapse Time-Based Data Archival

Metadata-Driven Hot/Cold Tier Archival Framework | Vivek Billore

25%+

Storage Reclaimed

70%

Manual Effort Reduced

Zero

Report Changes

Significant

Cost Savings

■ Problem Statement

Enterprise data warehouses accumulate billions of rows in fact and transaction tables over time. Without a retention strategy: query performance degrades, Synapse storage costs increase, manual archival is error-prone, and downstream reports break if tables are simply deleted. This framework solves all four with a **zero-disruption, metadata-driven archival pipeline**.

■ Architecture Flow

Archival_Metadata Table

Admin inserts ONE row to onboard any table — no code changes needed



Azure Data Factory Pipeline

PL_DATA_TIME_BASED_ARCHIVAL reads metadata, loops per table



CETAS Export to ADLS
Parquet

Aged rows exported to Hot or Cold tier based on config



External Table + UNION ALL
View

Hot tier exposed via External Table; View merges active + archived data



Delete from Synapse Table

Archived rows removed — storage reclaimed, queries faster



Audit Log Entry

Row counts, file paths, timestamps logged to Archival_AuditLog



■ ADF Pipeline Structure

Activity	Description
PL_DATA_TIME_BASED_ARCHIVAL	Master Pipeline — orchestrates all activities
ACT_LKP_GetMetadata	Read active rows from Archival_Metadata
ACT_FOREACH_Table	Loop over each table config row
ACT_COPY_ExportParquet	CETAS: export aged rows to ADLS as Parquet
ACT_SP_CreateExtTable	Create External Table (Hot tier only)
ACT_SP_CreateView	Create/update UNION ALL View

ACT_SP_DeleteArchived	Delete archived rows from physical table
ACT_SP_WriteAuditLog	Log run details to Archival_AuditLog

■ Hot vs Cold Tier

Feature	Hot Tier	Cold Tier
Storage	ADLS Standard	ADLS Archive (cheap)
Queryable	Yes — via External Table	No — stored only
View Merge	UNION ALL with physical table	Not exposed to SQL engine
Use Case	Historical data still in reports	Compliance / legal retention

■ SQL Objects Created Per Table

Object	Name Pattern	Purpose
External Table	ext_{TableName}_archived	Points to Hot tier Parquet in ADLS
UNION ALL View	vw_{TableName}	Merges physical + archived — zero report changes
Audit Log Row	Archival_AuditLog	Row counts, file paths, run status per execution

■ Tech Stack

Component	Technology
Source	Azure Synapse Analytics (Dedicated SQL Pool)
Orchestration	Azure Data Factory
Storage	ADLS Gen2 (Hot + Cold tiers)
File Format	Apache Parquet
Query Layer	Synapse External Tables + UNION ALL Views
Audit	SQL-based Archival_AuditLog table
Validation	PySpark row count reconciliation notebook

■ Core Design Principles

- Metadata-driven — no code changes to onboard new tables; just INSERT a config row
- Zero-disruption — UNION ALL Views preserve downstream query compatibility
- Hot/Cold separation — queryable vs compliance-only storage tiers
- Full auditability — every run logged with row counts, paths, timestamps
- Idempotent — IF NOT EXISTS checks make re-runs safe